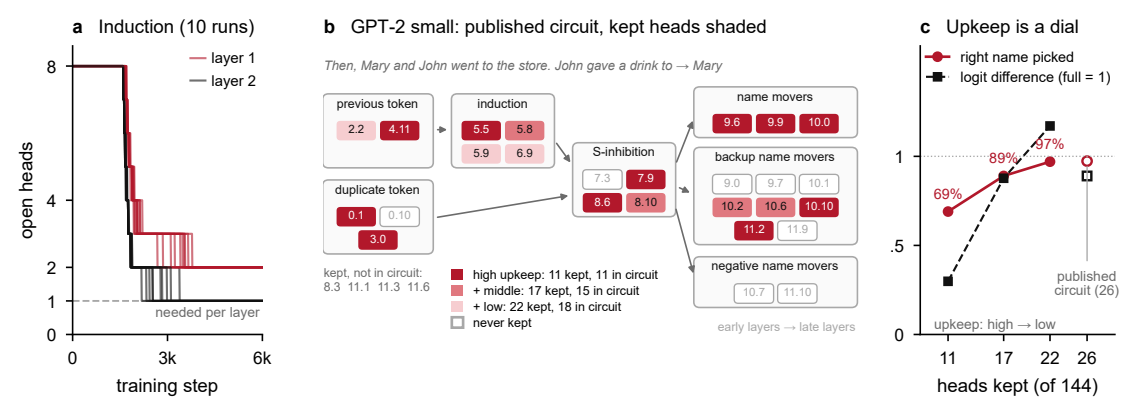

a Induction (10 runs)
layer 1
layer 2
open heads
8
4
2
1
0
needed per layer
0
3k
6k
training step
b GPT-2 small: published circuit, kept heads shaded
Then, Mary and John went to the store. John gave a drink to → Mary
previous token
2.2
4.11
induction
5.5
5.8
5.9
6.9
duplicate token
0.1
0.10
3.0
S-inhibition
7.3
7.9
8.6
8.10
name movers
9.6
9.9
10.0
backup name movers
9.0
9.7
10.1
10.2
10.6
10.10
11.2
11.9
negative name movers
10.7
11.10
kept, not in circuit:
8.3 11.1 11.3 11.6
high upkeep: 11 kept, 11 in circuit
+ middle: 17 kept, 15 in circuit
+ low: 22 kept, 18 in circuit
never kept
early layers → late layers
c Upkeep is a dial
right name picked
logit difference (full = 1)
1
.5
0
69%
89%
97%
published circuit (26)
upkeep: high → low
11
17
22
26
heads kept (of 144)